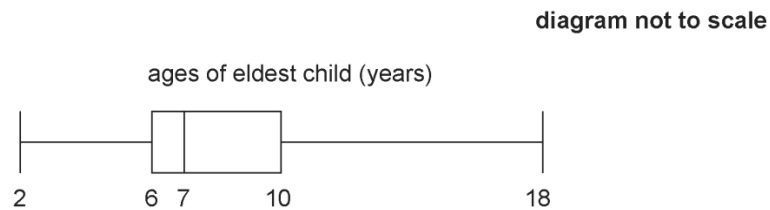


7-(MathAA/11_HL_Summer_2022_Q2)-Statistics

[Maximum mark: 7]

A survey at a swimming pool is given to one adult in each family. The age of the adult, a years old, and of their eldest child, c years old, are recorded.

The ages of the eldest child are summarized in the following box and whisker diagram.



- (a) Find the largest value of c that would not be considered an outlier. [3]

The regression line of a on c is $a = \frac{7}{4}c + 20$. The regression line of c on a is $c = \frac{1}{2}a - 9$.

- (b) (i) One of the adults surveyed is 42 years old. Estimate the age of their eldest child.
(ii) Find the mean age of all the adults surveyed. [4]



1-(MAT/12_HL_Summer_2012_Q3)-Probability

On a particular day, the probability that it rains is $\frac{2}{5}$. The probability that the “Tigers” soccer team wins on a day when it rains is $\frac{2}{7}$ and the probability that they win on a day when it does not rain is $\frac{4}{7}$.

- (a) Draw a tree diagram to represent these events and their outcomes.

[1 mark]

- (b) What is the probability that the “Tigers” soccer team wins?

[2 marks]

- (c) Given that the “Tigers” soccer team won, what is the probability that it rained on that day?

[2 marks]



2-(MAT/12_HL_Summer_2012_Q11)-Functions-Roots,Probability

Consider the following functions:

$$f(x) = \frac{2x^2 + 3}{75}, x \geq 0$$

$$g(x) = \frac{|3x - 4|}{10}, x \in \mathbb{R}.$$

- (a) State the range of f and of g . [2 marks]

- (b) Find an expression for the composite function $f \circ g(x)$ in the form $\frac{ax^2 + bx + c}{3750}$, where a , b and $c \in \mathbb{Z}$. [4 marks]

- (c) (i) Find an expression for the inverse function $f^{-1}(x)$.

- (ii) State the domain and range of f^{-1} . [4 marks]

The domains of f and g are now restricted to $\{0, 1, 2, 3, 4\}$.

- (d) By considering the values of f and g on this new domain, determine which of f and g could be used to find a probability distribution for a discrete random variable X , stating your reasons clearly. [6 marks]

- (e) Using this probability distribution, calculate the mean of X . [2 marks]



3-(MAT/10_HL_Winter_2012_Q5)-Probability

The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} ae^{-x}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

(a) State the mode of X .

[1 mark]

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(b) Determine the value of a .

[3 marks]

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(c) Find $E(X)$.

[4 marks]

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4-(MAT/11_HL_Summer_2013_Q4)-Probability

[Maximum mark: 5]

The probability density function of the random variable X is defined as

$$f(x) = \begin{cases} \sin x, & 0 \leq x \leq \frac{\pi}{2} \\ 0, & \text{otherwise.} \end{cases}$$

Find $E(X)$.

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5-(MAT/12_HL_Summer_2013_Q4)-Probability

Tim and Caz buy a box of 16 chocolates of which 10 are milk and 6 are dark. Caz randomly takes a chocolate and eats it. Then Tim randomly takes a chocolate and eats it.

- (a) Draw a tree diagram representing the possible outcomes, clearly labelling each branch with the correct probability.

[3 marks]

- (b) Find the probability that Tim and Caz eat the same type of chocolate.

[2 marks]

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6-(MAT/11_HL_Summer_2013_Q9)-Probability

[Maximum mark: 6]

Two events A and B are such that $P(A \cup B) = 0.7$ and $P(A|B') = 0.6$.

Find $P(B)$.

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7-(MAT/11_HL_Summer_2013_Q13)-Probability, Binomial Theorem

On Saturday, Alfred and Beatrice play 6 different games against each other. In each game, one of the two wins. The probability that Alfred wins any one of these games is $\frac{2}{3}$.

(a) Show that the probability that Alfred wins exactly 4 of the games is $\frac{80}{243}$. [3 marks]

(b) (i) Explain why the total number of possible outcomes for the results of the 6 games is 64.

(ii) By expanding $(1+x)^6$ and choosing a suitable value for x , prove

$$64 = \binom{6}{0} + \binom{6}{1} + \binom{6}{2} + \binom{6}{3} + \binom{6}{4} + \binom{6}{5} + \binom{6}{6}.$$

(iii) State the meaning of this equality in the context of the 6 games played. [4 marks]

(c) The following day Alfred and Beatrice play the 6 games again. Assume that the probability that Alfred wins any one of these games is still $\frac{2}{3}$.

(i) Find an expression for the probability Alfred wins 4 games on the first day and 2 on the second day. Give your answer in the form

$$\binom{6}{r} \left(\frac{2}{3}\right)^s \left(\frac{1}{3}\right)^t \text{ where the values of } r, s \text{ and } t \text{ are to be found.}$$

(ii) Using your answer to (c)(i) and 6 similar expressions write down the probability that Alfred wins a total of 6 games over the two days as the sum of 7 probabilities.

(iii) Hence prove that $\binom{12}{6} = \binom{6}{0}^2 + \binom{6}{1}^2 + \binom{6}{2}^2 + \binom{6}{3}^2 + \binom{6}{4}^2 + \binom{6}{5}^2 + \binom{6}{6}^2$. [9 marks]

(d) Alfred and Beatrice play n games. Let A denote the number of games Alfred wins.

The expected value of A can be written as $E(A) = \sum_{r=0}^n r \binom{n}{r} \frac{a^r}{b^n}$.

(i) Find the values of a and b .

(ii) By differentiating the expansion of $(1+x)^n$, prove that the expected number of games Alfred wins is $\frac{2n}{3}$. [6 marks]



8-(MAT/10_HL_Winter_2013_Q2)-Probability

The discrete random variable X has probability distribution:

x	0	1	2	3
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	a

- (a) Find the value of a . [1]
- (b) Find $E(X)$. [2]
- (c) Find $\text{Var}(X)$. [3]

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9-(MAT/12_HL_Summer_2014_Q1)-Probability

Events A and B are such that $P(A) = \frac{2}{5}$, $P(B) = \frac{11}{20}$ and $P(A|B) = \frac{2}{11}$.

- (a) Find $P(A \cap B)$. [2]
- (b) Find $P(A \cup B)$. [2]
- (c) State with a reason whether or not events A and B are independent. [2]

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10-(MAT/12_HL_Summer_2014_Q11)-Probability

- (a) Mobile phone batteries are produced by two machines. Machine A produces 60% of the daily output and machine B produces 40%. It is found by testing that on average 2% of batteries produced by machine A are faulty and 1% of batteries produced by machine B are faulty.

- (i) Draw a tree diagram clearly showing the respective probabilities.
- (ii) A battery is selected at random. Find the probability that it is faulty.
- (iii) A battery is selected at random and found to be faulty. Find the probability that it was produced by machine A. [6]

- (b) In a pack of seven transistors, three are found to be defective. Three transistors are selected from the pack at random without replacement. The discrete random variable X represents the number of defective transistors selected.

- (i) Find $P(X = 2)$.
- (ii) **Copy** and complete the following table:

x	0	1	2	3
$P(X = x)$				

- (iii) Determine $E(X)$. [6]



11-(MAT/10_HL_Winter_2014_Q4)-Probability

Events A and B are such that $P(A) = 0.2$ and $P(B) = 0.5$.

(a) Determine the value of $P(A \cup B)$ when

(i) A and B are mutually exclusive;

(ii) A and B are independent.

[4]

(b) Determine the range of possible values of $P(A|B)$.

[3]

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12-(MAT/10_HL_Winter_2014_Q9)-Probability

A continuous random variable T has probability density function f defined by

$$f(t) = \begin{cases} |2-t|, & 1 \leq t \leq 3 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Sketch the graph of $y = f(t)$. [2]
- (b) Find the interquartile range of T . [4]

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13-(MAT/12_HL_Summer_2015_Q1)-Probability

A and B are two events such that $P(A) = 0.25$, $P(B) = 0.6$ and $P(A \cup B) = 0.7$.

(a) Find $P(A \cap B)$. [2]

(b) Determine whether events A and B are independent. [2]

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14-(MAT/11_HL_Summer_2015_Q10)-Probability

[Maximum mark: 6]

A football team, Melchester Rovers are playing a tournament of five matches.

The probabilities that they win, draw or lose a match are $\frac{1}{2}$, $\frac{1}{6}$ and $\frac{1}{3}$ respectively.

These probabilities remain constant; the result of a match is independent of the results of other matches. At the end of the tournament their coach Roy loses his job if they lose three **consecutive** matches, otherwise he does not lose his job. Find the probability that Roy loses his job.

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15-(MAT/10_HL_Winter_2015_Q6)-Probability

A box contains four red balls and two white balls. Darren and Marty play a game by each taking it in turn to take a ball from the box, without replacement. The first player to take a white ball is the winner.

- (a) Darren plays first, find the probability that he wins. [4]

The game is now changed so that the ball chosen is replaced after each turn. Darren still plays first.

- (b) Show that the probability of Darren winning has not changed. [3]

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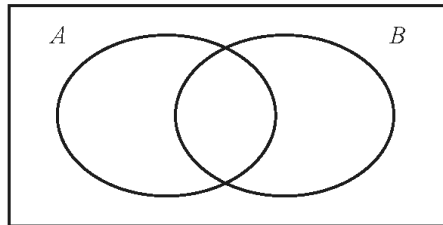
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16-(MAT/11_HL_Summer_2016_Q4)-Probability

- (a) On the Venn diagram shade the region $A' \cap B'$. [1]



Two events A and B are such that $P(A \cap B') = 0.2$ and $P(A \cup B) = 0.9$.

- (b) Find $P(A' | B')$. [4]

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17-(MAT/12_HL_Summer_2016_Q5)-Probability

A biased coin is tossed five times. The probability of obtaining a head in any one throw is p .

Let X be the number of heads obtained.

- (a) Find, in terms of p , an expression for $P(X = 4)$. [2]
- (b) (i) Determine the value of p for which $P(X = 4)$ is a maximum.
- (ii) For this value of p , determine the expected number of heads. [6]

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18-(MAT/12_HL_Summer_2016_Q7)-Probability

A and B are independent events such that $P(A) = P(B) = p$, $p \neq 0$.

(a) Show that $P(A \cup B) = 2p - p^2$. [2]

(b) Find $P(A | A \cup B)$ in simplest form. [4]

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19–(MAT/10_HL_Winter_2016_Q2)–Probability

The faces of a fair six-sided die are numbered 1, 2, 2, 4, 4, 6. Let X be the discrete random variable that models the score obtained when this die is rolled.

- (a) Complete the probability distribution table for X . [2]

x				
$P(X = x)$				

- (b) Find the expected value of X . [2]



20-(MAT/10_HL_Winter_2016_Q10)-Probability

Consider two events A and B defined in the same sample space.

(a) Show that $P(A \cup B) = P(A) + P(A' \cap B)$. [3]

(b) Given that $P(A \cup B) = \frac{4}{9}$, $P(B | A) = \frac{1}{3}$ and $P(B | A') = \frac{1}{6}$,

(i) show that $P(A) = \frac{1}{3}$;

(ii) hence find $P(B)$. [6]



21-(MAT/11_HL_Summer_2017_Q4)-Probability

[Maximum mark: 5]

Three girls and four boys are seated randomly on a straight bench. Find the probability that the girls sit together and the boys sit together.



22-(MAT/12_HL_Summer_2017_Q7)-Statistics, Probability

- (a) The random variable X has the Poisson distribution $\text{Po}(m)$. Given that $P(X > 0) = \frac{3}{4}$, find the value of m in the form $\ln a$ where a is an integer. [3]
- (b) The random variable Y has the Poisson distribution $\text{Po}(2m)$. Find $P(Y > 1)$ in the form $\frac{b - \ln c}{c}$ where b and c are integers. [4]



23-(MAT/11_HL_Summer_2017_Q10)-Probability, Statistics

The continuous random variable X has a probability density function given by

$$f(x) = \begin{cases} k \sin\left(\frac{\pi x}{6}\right), & 0 \leq x \leq 6 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the value of k . [4]
- (b) By considering the graph of f write down
- (i) the mean of X ;
 - (ii) the median of X ;
 - (iii) the mode of X . [3]
- (c) (i) Show that $P(0 \leq X \leq 2) = \frac{1}{4}$.
- (ii) Hence state the interquartile range of X . [6]
- (d) Calculate $P(X \leq 4 \mid X \geq 3)$. [2]



24–(MAT/11_HL_Summer_2018_Q3)–Probability

Two unbiased tetrahedral (four-sided) dice with faces labelled 1, 2, 3, 4 are thrown and the scores recorded. Let the random variable T be the maximum of these two scores. The probability distribution of T is given in the following table.

t	1	2	3	4
$P(T = t)$	$\frac{1}{16}$	a	b	$\frac{7}{16}$

- (a) Find the value of a and the value of b . [3]
- (b) Find the expected value of T . [2]



25-(MAT/12_HL_Summer_2018_Q3)-Probability

The discrete random variable X has the following probability distribution, where p is a constant.

x	0	1	2	3	4
$P(X = x)$	p	$0.5 - p$	0.25	0.125	p^3

(a) Find the value of p . [2]

(b) (i) Find μ , the expected value of X .

(ii) Find $P(X > \mu)$. [4]



26–(MAT/10_HL_Winter_2018_Q1)–Probability

Consider two events, A and B , such that $P(A) = P(A' \cap B) = 0.4$ and $P(A \cap B) = 0.1$.

- (a) By drawing a Venn diagram, or otherwise, find $P(A \cup B)$. [3]
- (b) Show that the events A and B are not independent. [3]



27–(MAT/11_HL_Summer_2019_Q6)–Probability

Let X be a random variable which follows a normal distribution with mean μ . Given that $P(X < \mu - 5) = 0.2$, find

- (a) $P(X > \mu + 5)$: [2]
- (b) $P(X < \mu + 5 \mid X > \mu - 5)$. [5]



28–(MAT/11_HL_Summer_2019_Q10)–Remainder&FactorTheorem,Probability

The function $p(x)$ is defined by $p(x) = x^3 - 3x^2 + 8x - 24$ where $x \in \mathbb{R}$.

(a) Find the remainder when $p(x)$ is divided by

(i) $(x - 2)$

(ii) $(x - 3)$.

[3]

(b) Prove that $p(x)$ has only one real zero.

[4]

(c) Write down the transformation that will transform the graph of $y = p(x)$ onto the graph of $y = 8x^3 - 12x^2 + 16x - 24$.

[2]

The random variable X follows a Poisson distribution with a mean of λ and $6P(X = 3) = 3P(X = 2) - 2P(X = 1) + 3P(X = 0)$.

(d) Find the value of λ .

[6]



29–(MAT/12_HL_Summer_2019_Q10)–Integration, Probability

The random variable X has probability density function f given by

$$f(x) = \begin{cases} k(\pi - \arcsin x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}, \text{ where } k \text{ is a positive constant.}$$

(a) State the mode of X . [1]

(b) (i) Find $\int \arcsin x \, dx$.

(ii) Hence show that $k = \frac{2}{2 + \pi}$. [6]

(c) Given that $y = \left(\frac{x^2}{2}\right)\arcsin x - \left(\frac{1}{4}\right)\arcsin x + \left(\frac{x}{4}\right)\sqrt{1-x^2}$, show that

(i) $\frac{dy}{dx} = x \arcsin x$;

(ii) $E(X) = \frac{3\pi}{4(\pi+2)}$. [9]



30-(MAT/10_HL_Winter_2019_Q1)-Probability

[Maximum mark: 5]

The probability distribution of a discrete random variable, X , is given by the following table, where N and p are constants.

x	1	5	10	N
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{5}$	$\frac{1}{5}$	p

- (a) Find the value of p . [2]
- (b) Given that $E(X) = 10$, find the value of N . [3]

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31–(MAT/10_HL_Winter_2020_Q1)–Probability

[Maximum mark: 4]

A discrete random variable X has the probability distribution given by the following table.

x	0	1	2	3
$P(X=x)$	p	$\frac{1}{4}$	$\frac{1}{6}$	q

Given that $E(X) = \frac{19}{12}$, determine the value of p and the value of q .

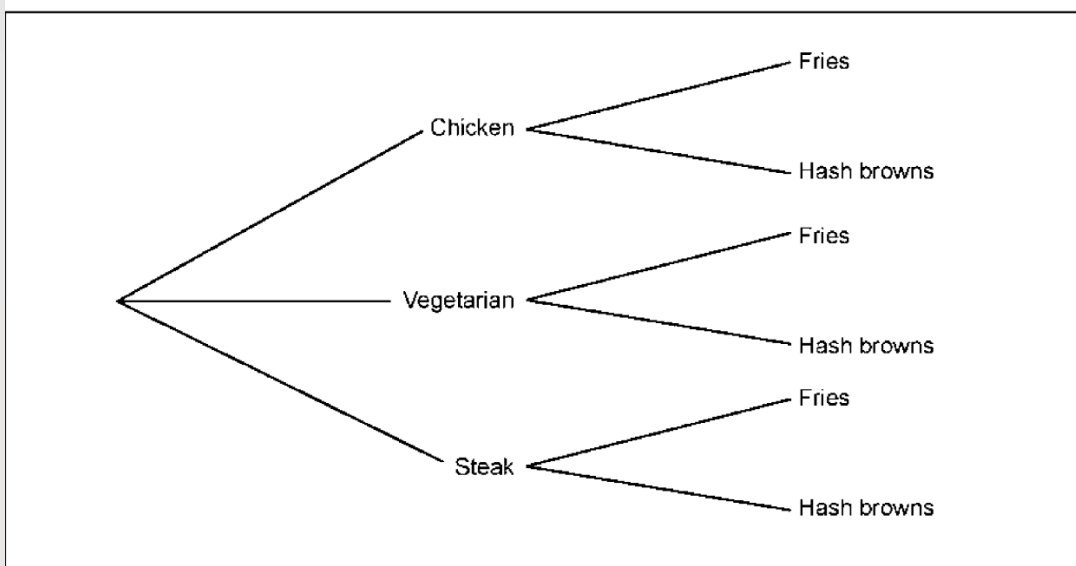


32-(MAT/10_HL_Winter_2020_Q3)-Probability

At Nusaybah's Breakfast Diner, three types of omelette are available to order: chicken, vegetarian and steak. Each omelette is served with either a portion of fries or hash browns. It is known that 20% of customers choose a chicken omelette, 70% choose a vegetarian omelette and 10% choose a steak omelette.

It is also known that 65% of those ordering the chicken omelette, 70% of those ordering the vegetarian omelette and 60% of those ordering the steak omelette, order fries.

The following tree diagram represents the orders made by each customer.



- (a) Complete the tree diagram by adding the respective probabilities to each branch. [2]
- (b) Find the probability that a randomly selected customer orders fries. [2]
- (c) Find the probability that a randomly selected customer orders fries, given that they do **not** order a chicken omelette. [3]



33-(MathAA/11_HL_Summer_2021_Q10)-Probability

[Maximum mark: 16]

A biased four-sided die, A, is rolled. Let X be the score obtained when die A is rolled. The probability distribution for X is given in the following table.

x	1	2	3	4
$P(X=x)$	p	p	p	$\frac{1}{2}p$

(a) Find the value of p . [2]

(b) Hence, find the value of $E(X)$. [2]

A second biased four-sided die, B, is rolled. Let Y be the score obtained when die B is rolled. The probability distribution for Y is given in the following table.

y	1	2	3	4
$P(Y=y)$	q	q	q	r

(c) (i) State the range of possible values of r .

(ii) Hence, find the range of possible values of q . [3]

(d) Hence, find the range of possible values for $E(Y)$. [3]

Agnes and Barbara play a game using these dice. Agnes rolls die A once and Barbara rolls die B once. The probability that Agnes' score is less than Barbara's score is $\frac{1}{2}$.

(e) Find the value of $E(Y)$. [6]



34–(MathAA/10_HL_Winter_2021_Q4)–Probability

[Maximum mark: 5]

Box 1 contains 5 red balls and 2 white balls.

Box 2 contains 4 red balls and 3 white balls.

- (a) A box is chosen at random and a ball is drawn. Find the probability that the ball is red. [3]

Let A be the event that “box 1 is chosen” and let R be the event that “a red ball is drawn”.

- (b) Determine whether events A and R are independent. [2]



35–(MathAA/11_HL_Summer_2022_Q7)–Probability

[Maximum mark: 8]

The continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{k}{\sqrt{4-3x^2}}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the value of k . [4]
- (b) Find $E(X)$. [4]



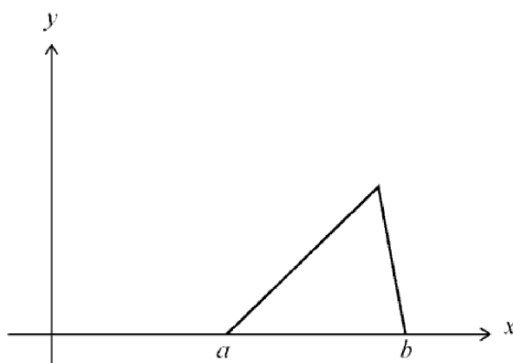
36–(MathAA/12_HL_Summer_2022_Q8)–Probability

[Maximum mark: 6]

A continuous random variable X has the probability density function

$$f(x) = \begin{cases} \frac{2}{(b-a)(c-a)}(x-a), & a \leq x \leq c \\ \frac{2}{(b-a)(b-c)}(b-x), & c < x \leq b \\ 0, & \text{otherwise} \end{cases}$$

The following diagram shows the graph of $y = f(x)$ for $a \leq x \leq b$.



Given that $c \geq \frac{a+b}{2}$, find an expression for the median of X in terms of a , b and c .



37-(MathAA/12_HL_Summer_2022_Q10)-Probability

[Maximum mark: 16]

A biased four-sided die with faces labelled 1, 2, 3 and 4 is rolled and the result recorded. Let X be the result obtained when the die is rolled. The probability distribution for X is given in the following table where p and q are constants.

x	1	2	3	4
$P(X = x)$	p	0.3	q	0.1

For this probability distribution, it is known that $E(X) = 2$.

- (a) Show that $p = 0.4$ and $q = 0.2$. [5]
- (b) Find $P(X > 2)$. [2]

Nicky plays a game with this four-sided die. In this game she is allowed a maximum of five rolls. Her score is calculated by adding the results of each roll. Nicky wins the game if her score is at least ten.

After three rolls of the die, Nicky has a score of four.

- (c) Assuming that rolls of the die are independent, find the probability that Nicky wins the game. [5]

David has two pairs of unbiased four-sided dice, a yellow pair and a red pair. Both yellow dice have faces labelled 1, 2, 3 and 4. Let S represent the sum obtained by rolling the two yellow dice. The probability distribution for S is shown below.

s	2	3	4	5	6	7	8
$P(S = s)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

The first red die has faces labelled 1, 2, 2 and 3. The second red die has faces labelled 1, a , a and b , where $a < b$ and $a, b \in \mathbb{Z}^+$. The probability distribution for the sum obtained by rolling the red pair is the same as the distribution for the sum obtained by rolling the yellow pair.

- (d) Determine the value of b . [2]
- (e) Find the value of a , providing evidence for your answer. [2]

